

Pearson Edexcel International GCSE

Information and Communication Technology (ICT)

Paper 1: Written Paper (Specialist Focus: Cloud Computing)

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Total Marks: 100 Marks

Instructions to Candidates

- Use black ink or ball-point pen. Fill in your name, centre number, and candidate number safely.
- Answer all questions in the provided organized response spaces block panels.
- Cross out any mistakes cleanly; do not use correction fluid or messy erasures.

Information for Candidates

- The total mark for this exam paper is 100. Marks per question are detailed in brackets ().
- This paper has been optimized to evaluate direct recall, core definitions, and technical scenarios.

SECTION A: Digital Devices and Terminology (25 Marks)

Question 1

A student uses an electronic device to connect to a school network database.

(a) State what is meant by the term "local hardware". (1 Mark)

Definition of local hardware...

(b) State what is meant by the term "the cloud". (1 Mark)

Definition of the cloud...

(c) Identify one example of an online application that users can use to store or sync files. (1 Mark)

Example application...

(d) State two types of consumer devices that can be used to connect to cloud networks. (2 Marks)

1. 2.

Question 2

The growth of cloud services has influenced the manufacturing of low-cost digital hardware devices.

(a) Identify two types of compact, low-specification computers whose manufacturing has increased due to cloud technology. (2 Marks)

1. 2.

(b) State two internal hardware components that do not need to be large or ultra-fast on devices using cloud processing. (2 Marks)

1. 2.

(c) Define the term "netbook" using the vocabulary guidelines. (2 Marks)

Definition of netbook...

Question 3 (Multiple Choice)

Select the correct response for each question below. Place a cross [X] inside the accurate checkbox container.

(a) Which one of these devices is primarily used to store and execute hosted application programs? (1 Mark)

☐ A) Laptop

☐ B) Smartwatch

☐ C) Server

☐ D) Tablet

(b) What does the term "edit" mean in relation to working with documents? (1 Mark)

☐ A) Deleting a file completely from an online storage network folder.

☐ B) A modification or change made to a document.

☐ C) Copying data to an external physical storage medium.

☐ D) Restricting a user's permission to view a file.

Question 4

Match the following vocabulary definitions by selecting the correct term from the list.

Vocabulary List: *Application Loading Time, Unresponsive, Scalable*

(a) Not reacting to user input or commands, often caused by lag. (1 Mark)

Matched term...

(b) The capacity of a system (like storage or processing) to grow or shrink flexibly based on demand. (1 Mark)

Matched term...

(c) The total time it takes for an application to open and become ready for use. (1 Mark)

Matched term...

Question 5

Complete the sentences below about cloud networks by entering the missing core words.

(a) Web applications (web apps)—also known as _____—are programs stored and run on online servers. (1 Mark)

Missing word(s)...

(b) The user interface of an online hosted application is displayed to you through a standard web _____. (1 Mark)

Missing word...

(c) Hosted applications rely on a network interaction configuration called the _____ model. (1 Mark)

Missing model type...

(d) Working locally on a computer's hard drive generally provides the best performance speed because there is no _____ required to transfer data. (1 Mark)

Missing parameter...

SECTION B: Connectivity and Performance Factors (25 Marks)

Question 6

Hosted software solutions are only available through an active connection to the internet.

(a) State the two network metrics that heavily influence the performance of hosted applications. (2 Marks)

Metric 1: Metric 2:

(b) State two distinct user experience drops that occur when a connection suffers from low bandwidth or high latency. (2 Marks)

Drop 1: Drop 2:

Question 7

Hosted application developers design software features differently to help optimize performance.

(a) State how the complexity of features in a hosted application compares to locally installed software. (1 Mark)

Comparison statement...

(b) Explain how reducing the amount of data processed keeps the user experience smoother when a connection is slow. (2 Marks)

Explanation mechanism...

Question 8

Cloud systems allow groups of individuals to work together on projects from separate physical spaces.

(a) State the term used to describe multiple users modifying a document at the exact same time. (1 Mark)

Technical term...

(b) Identify two collaborative features built into hosted application platforms that help teams manage group document edits. (2 Marks)

Feature 1: Feature 2:

Question 9

State whether each statement is True or False based on network connection parameters.

(a) Hosted software applications do not require an active internet connection to function. (1 Mark)

True or False...

(b) Working with files directly on your computer's hard drive generally provides better performance speed than cloud transfers. (1 Mark)

True or False...

(c) High latency makes an application load faster and feel more responsive. (1 Mark)

True or False...

SECTION C: Benefits, Drawbacks, and Storage Management (30 Marks)

Question 10

Complete the evaluation grid by checking [X] to classify each statement feature as a Benefit or a Drawback. (5 Marks)

Feature of Hosted Applications	Benefit [X]	Drawback [X]
Accessible from many types of connected devices.	☐	☐
Only accessible when you have an active internet connection.	☐	☐
Generally has fewer advanced features compared to local software.	☐	☐
Includes integrated online data storage to save files online.	☐	☐
Features powerful built-in tools that enable group collaboration.	☐	☐

Question 11

An individual chooses to use an online data storage provider instead of upgrading their physical hardware memory drives.

(a) State three distinct benefits of using online data storage services. (3 Marks)

Benefit 1: Benefit 2: Benefit 3:

(b) State three distinct drawbacks of using online data storage services. (3 Marks)

Drawback 1: Drawback 2: Drawback 3:

Question 12

Data management parameters differ between local drives and remote host systems.

(a) Compare the speed of transferring files using online cloud storage versus local storage. (1 Mark)

Comparison statement...

(b) State what happens to file management convenience when some cloud file formats cannot be opened directly online. (1 Mark)

Impact statement...

(c) Describe who is responsible for performing automated data backups when utilizing cloud storage platforms. (2 Marks)

Backup responsibility description...

Question 13

An organization is transferring its private document libraries to an online service vendor.

(a) Define the concept "data protection issues" based on standard regulatory frameworks. (2 Marks)

Definition...

(b) State what compliance risk arises when data files are stored on servers physically located in another country. (1 Mark)

Compliance cross-border risk...

(c) Explain why security features change when a business transitions file storage to an external cloud provider. (2 Marks)

Security change explanation...

SECTION D: Social and Organizational Impacts (20 Marks)

Question 14

Cloud technology alters the way organizations operate and how individuals interact with online networks daily.

(a) Identify three types of everyday online services used by individuals that are driven by cloud infrastructure. (3 Marks)

1. 2. 3.

(b) Identify three types of online interactive communities or platforms hosted by cloud infrastructure. (3 Marks)

1. 2. 3.

(c) State two massive physical overhead expenses that a business completely eliminates by switching from in-house physical servers to cloud infrastructure. (2 Marks)

1. 2.

Question 15

The accessibility of modern digital tools can impact socioeconomic divides between populations.

(a) State what is meant by the phrase "the digital divide". (2 Marks)

Definition of digital divide...

(b) Explain how lower hardware purchasing costs help more individuals access technology and reduce this divide. (2 Marks)

Socioeconomic access explanation...

SECTION E: Short Scenario Evaluation (8 Marks)

Question 16

A small community library wants to change its local word-processing software and physical storage flash drives to an online hosted web application suite with cloud data storage.

Discuss the advantages and disadvantages for the library if it moves to this cloud system. In your response, ensure you mention **access flexibility**, **team collaboration tools**, **internet connection dependencies**, and **monthly space subscription costs**. (8 Marks)

Provide a balanced structural evaluation essay answer here...

End of Question Bank